

Lab 06 NAT

Note: I used the trace file in the lab Doc and didnt do extra credit.

1. NAT Measurement Scenario

1. What is the IP address of the client?

The client address is 192.168.1.100

56	13:43:07.378402	192.168.1.100	64.233.169.104	HTTP	689	GET / HTTP/1.1
57	13:43:07.409863	64.233.169.104	192.168.1.100	TCP	60	80 → 4335 [ACK] Seq=1 Ack=636 Wi
58	13:43:07.427567	64.233.169.104	192.168.1.100	TCP	1484	80 → 4335 [ACK] Seq=1 Ack=636 Wi
59	13:43:07.427896	64.233.169.104	192.168.1.100	TCP	1484	80 → 4335 [ACK] Seq=1431 Ack=636
60	13:43:07.427932	64.233.169.104	192.168.1.100	HTTP	814	HTTP/1.1 200 OK (text/html)
61	13:43:07.427979	192.168.1.100	64.233.169.104	TCP	54	4335 → 80 [ACK] Seq=636 Ack=3621
62	13:43:07.550534	192.168.1.100	64.233.169.104	HTTP	719	GET /intl/en_ALL/images/logo.gif

2. The client actually communicates with several different Google servers in order to implement “safe browsing.” (See extra credit section at the end of this lab). The main Google server that will serve up the main Google web page has IP address 64.233.169.104. In order to display only those frames containing HTTP messages that are sent to/from this Google, server, enter the expression “http && ip.addr == 64.233.169.104” (without quotes) into the Filter: field in Wireshark).

The image shows a Wireshark capture of a network trace. The filter bar at the top contains the expression `http && ip.addr == 64.233.169.104`. The packet list shows several HTTP GET requests from the client (192.168.1.100) to the Google server (64.233.169.104). The packet details pane shows the structure of the selected HTTP packet, including Ethernet II, Internet Protocol Version 4, Transmission Control Protocol, and Hypertext Transfer Protocol.

No.	Time	Source	Destination	Protocol	Length	Info
56	13:43:07.378402	192.168.1.100	64.233.169.104	HTTP	689	GET / HTTP/1.1
60	13:43:07.427932	64.233.169.104	192.168.1.100	HTTP	814	HTTP/1.1 200 OK (text/html)
62	13:43:07.550534	192.168.1.100	64.233.169.104	HTTP	719	GET /intl/en_ALL/images/logo.gif HTTP
73	13:43:07.618586	64.233.169.104	192.168.1.100	HTTP	226	HTTP/1.1 200 OK (GIF89a)
75	13:43:07.639320	192.168.1.100	64.233.169.104	HTTP	809	GET /extern_js/f/CgJlbhICdXMrMAo4NUA
92	13:43:07.717784	64.233.169.104	192.168.1.100	HTTP	648	HTTP/1.1 200 OK (text/javascript)
94	13:43:07.761459	192.168.1.100	64.233.169.104	HTTP	695	GET /extern_chrome/ee36edbd3c16a1c5.
100	13:43:07.806488	64.233.169.104	192.168.1.100	HTTP	870	HTTP/1.1 200 OK (text/html)
107	13:43:07.921971	192.168.1.100	64.233.169.104	HTTP	712	GET /images/nav_logo7.png HTTP/1.1
112	13:43:07.951496	192.168.1.100	64.233.169.104	HTTP	806	GET /csi?v=3&s=webhp&action=&tran=un
119	13:43:07.954921	64.233.169.104	192.168.1.100	HTTP	1359	HTTP/1.1 200 OK (PNG)
122	13:43:07.978625	192.168.1.100	64.233.169.104	HTTP	670	GET /favicon.ico HTTP/1.1
124	13:43:08.006918	64.233.169.104	192.168.1.100	HTTP	269	HTTP/1.1 204 No Content
127	13:43:08.032636	64.233.169.104	192.168.1.100	HTTP	1204	HTTP/1.1 200 OK (image/x-icon)

Frame 56: 689 bytes on wire (5512 bits), 689 bytes captured (5512 bits)
 Ethernet II, Src: HonHaiPrecis_0d:ca:8f (00:22:68:0d:ca:8f), Dst: CiscoLinksys_45:1f:1b (00:22:6b:45:1f:1b)
 Internet Protocol Version 4, Src: 192.168.1.100, Dst: 64.233.169.104
 Transmission Control Protocol, Src Port: 4335, Dst Port: 80, Seq: 1, Ack: 1, Len: 635
 Hypertext Transfer Protocol

3. Consider now the HTTP GET sent from the client to the Google server (whose IP address is IP address 64.233.169.104) at time 7.109267. What are the source and destination IP addresses and TCP source and destination ports on the IP datagram carrying this HTTP GET?

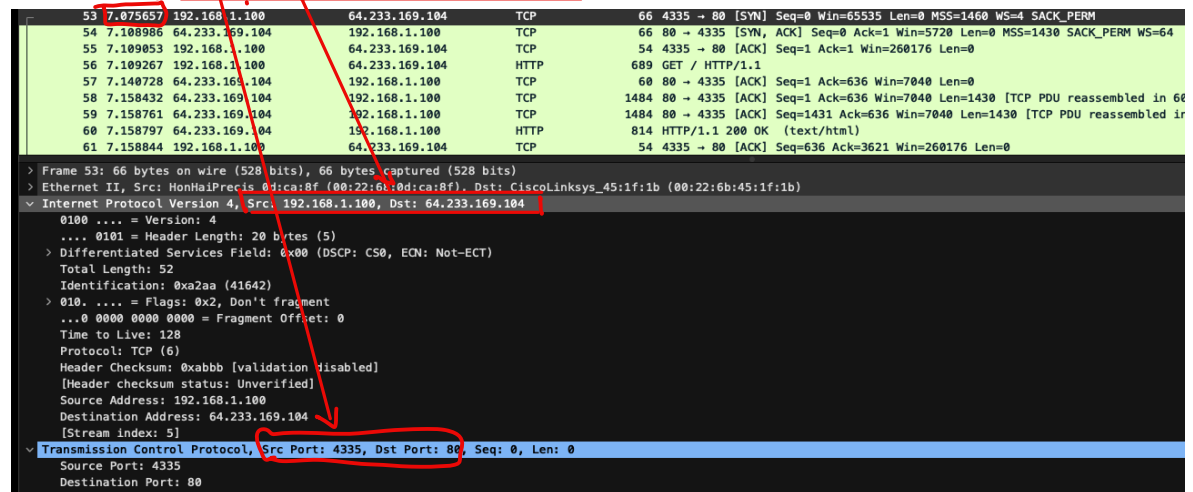
5. Recall that before a GET command can be sent to an HTTP server, TCP must first set up a connection using the three-way SYN/ACK handshake. At what time is the client-to-server TCP SYN segment sent that sets up the connection used by the GET sent at time 7.109267? What are the source and destination IP addresses and source and destination ports for the TCP SYN segment? What are the source and destination IP addresses and source and destination ports of the ACK sent in response to the SYN. At what time is this ACK received at the client? (Note: to find these segments you will need to clear the Filter expression you entered above in step 2. If you enter the filter "tcp", only TCP segments will be displayed by Wireshark).

SYN:

At the time 7.075657, this message appears in the trace file.

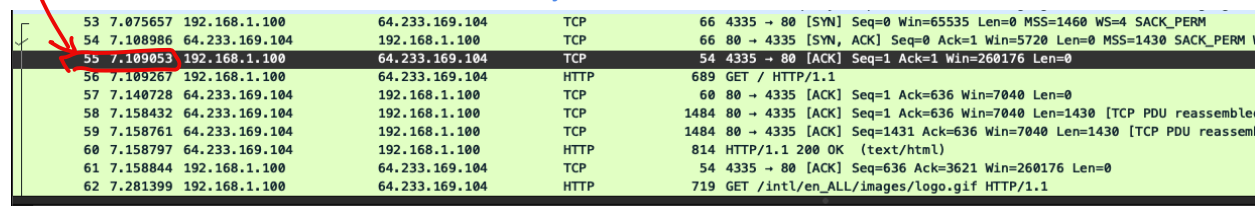
The source IP Src: 192.168.1.100, Dst: 64.233.169.104;

TCP: Src Port: 4335, Dst Port: 80



ACK:

At the time 7.109053, this ACK received by the client



IP Source: 64.233.169.104 Dest: 192.168.1.100

TCP: Source: 80 Dst: 4335

6. In the NAT_ISP_side trace file, find the HTTP GET message was sent from the client to the Google server at time 7.109267 (where $t=7.109267$ is time at which this was sent as recorded in the NAT_home_side trace file). At what time does this message appear in the NAT_ISP_side trace file? What are the source and destination IP addresses and TCP source and destination ports on the IP datagram carrying this HTTP GET (as recording in the NAT_ISP_side trace file)? Which of these fields are the same, and which are different, than in your answer to question 3 above?

At the same time, 6.069168 this message appeared in the trace file,
IP: Src: 71.192.34.104, Dst: 64.233.169.104

TCP: Src Port: 4335, Dst Port: 80

The Src port, the Dst port number, and the Dest IP address are the same. Only the source IP address is different.

No.	Time	Source	Destination	Protocol	Length	Info
80	6.020086	71.192.34.104	68.87.71.230	DNS	74	Standard query 0xed6a A www.google.com
81	6.032738	68.87.71.230	71.192.34.104	DNS	158	Standard query response 0xed6a A www.google.com CNAME www.l.google.com A
82	6.035475	71.192.34.104	64.233.169.104	TCP	66	4335 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=4 SACK_PERM
83	6.067175	64.233.169.104	71.192.34.104	TCP	66	80 → 4335 [SYN, ACK] Seq=0 Ack=1 Win=5720 Len=0 MSS=1430 SACK_PERM WS=64
84	6.068754	71.192.34.104	64.233.169.104	TCP	60	4335 → 80 [ACK] Seq=1 Ack=1 Win=260176 Len=0
85	6.069168	71.192.34.104	64.233.169.104	HTTP	689	GET / HTTP/1.1
86	6.092755	Cisco_bf:6c:01	Broadcast	ARP	60	Who has 71.192.35.144? Tell 71.192.32.1
87	6.099637	64.233.169.104	71.192.34.104	TCP	60	80 → 4335 [ACK] Seq=1 Ack=636 Win=7040 Len=0
88	6.117078	64.233.169.104	71.192.34.104	TCP	1484	80 → 4335 [ACK] Seq=1 Ack=636 Win=7040 Len=1430 [TCP PDU reassembled in
89	6.117407	64.233.169.104	71.192.34.104	TCP	1484	80 → 4335 [ACK] Seq=1431 Ack=636 Win=7040 Len=1430 [TCP PDU reassembled
90	6.117570	64.233.169.104	71.192.34.104	HTTP	814	HTTP/1.1 200 OK (text/html)
91	6.118515	71.192.34.104	64.233.169.104	TCP	60	4335 → 80 [ACK] Seq=636 Ack=3621 Win=260176 Len=0
92	6.162091	169.254.255.255	169.254.255.255	NBNS	92	Name query NB HPAB9D4C<00>
93	6.241357	71.192.34.104	64.233.169.104	HTTP	719	GET /intl/en_ALL/images/logo.gif HTTP/1.1
94	6.273849	64.233.169.104	71.192.34.104	TCP	309	80 → 4335 [PSH, ACK] Seq=3621 Ack=1301 Win=8320 Len=255 [TCP PDU reassembled
95	6.274230	64.233.169.104	71.192.34.104	TCP	1484	80 → 4335 [ACK] Seq=3876 Ack=1301 Win=8320 Len=1430 [TCP PDU reassembled

> Frame 85: 689 bytes on wire (5512 bits), 689 bytes captured (5512 bits) on 0

> Ethernet II, Src: Dell_4f:36:23 (00:08:74:4f:36:23), Dst: Cisco_bf:6c:01 (00:0e:d6:bf:6c:01)

> Internet Protocol Version 4, Src: 71.192.34.104, Dst: 64.233.169.104

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 675

Identification: 0xa2ac (41644)

> 010. = Flags: 0x2, Don't fragment

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 127

Protocol: TCP (6)

Header Checksum: 0x022f [validation disabled]

[Header checksum status: Unverified]

Source Address: 71.192.34.104

Destination Address: 64.233.169.104

[Stream index: 7]

> Transmission Control Protocol, Src Port: 4335, Dst Port: 80, Seq: 1, Ack: 1, Len: 635

Source Port: 4335

Destination Port: 80

7. Are any fields in the HTTP GET message changed? Which of the following fields in the IP datagram carrying the HTTP GET are changed: Version, Header Length, Flags, Checksum. If any of these fields have changed, give a reason (in one sentence) stating why this field needed to change.

There are no fields in the HTTP Get message changed. For Version, headerLength, and Flags are the same, which means no changes, the only field changed is the header checksum. The header checksum is changed because the source IP address is changed.

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53 7.075657 192.168.1.100 64.233.169.104 TCP 66 4335 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=4 SACK_PERM
54 7.108986 64.233.169.104 192.168.1.100 TCP 66 80 → 4335 [SYN, ACK] Seq=0 Ack=1 Win=5720 Len=0 MSS=1430 SACK_PERM WS=64
55 7.109053 192.168.1.100 64.233.169.104 TCP 54 4335 → 80 [ACK] Seq=1 Ack=1 Win=260176 Len=0
56 7.109267 192.168.1.100 64.233.169.104 HTTP 689 GET / HTTP/1.1
57 7.140728 64.233.169.104 192.168.1.100 TCP 60 80 → 4335 [ACK] Seq=1 Ack=636 Win=7040 Len=0
58 7.158432 64.233.169.104 192.168.1.100 TCP 1484 80 → 4335 [ACK] Seq=1 Ack=636 Win=7040 Len=1430 [TCP PDU reassembled in 60
59 7.158761 64.233.169.104 192.168.1.100 TCP 1484 80 → 4335 [ACK] Seq=1431 Ack=636 Win=7040 Len=1430 [TCP PDU reassembled in
60 7.158797 64.233.169.104 192.168.1.100 HTTP 814 HTTP/1.1 200 OK (text/html)
61 7.158844 192.168.1.100 64.233.169.104 TCP 54 4335 → 80 [ACK] Seq=636 Ack=3621 Win=260176 Len=0

> Frame 53: 66 bytes on wire (528 bits), 66 bytes captured (528 bits)
> Ethernet II, Src: HonHaiPrecis_0d:ca:8f (00:22:68:0d:ca:8f), Dst: CiscoLinksys_45:1f:1b (00:22:6b:45:1f:1b)
> Internet Protocol Version 4, Src: 192.168.1.100, Dst: 64.233.169.104
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 52
  Identification: 0xa2aa (41642)
  > 010. .... = Flags: 0x2, Don't fragment
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 128
  Protocol: TCP (6)
  Header Checksum: 0xabbl [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.1.100
  Destination Address: 64.233.169.104
  [Stream index: 5]
  > Transmission Control Protocol, Src Port: 4335, Dst Port: 80, Seq: 0, Len: 0
    Source Port: 4335
    Destination Port: 80
    [Stream index: 2]

85 6.069168 71.192.34.104 64.233.169.104 HTTP 689 GET / HTTP/1.1
86 6.092755 Cisco_bf:6c:01 Broadcast ARP 60 Who has 71.192.35.144? Tell 71.192.32.1
87 6.099637 64.233.169.104 71.192.34.104 TCP 60 80 → 4335 [ACK] Seq=1 Ack=636 Win=7040 Len=0
88 6.117078 64.233.169.104 71.192.34.104 TCP 1484 80 → 4335 [ACK] Seq=1 Ack=636 Win=7040 Len=1430
89 6.117407 64.233.169.104 71.192.34.104 TCP 1484 80 → 4335 [ACK] Seq=1431 Ack=636 Win=7040 Len=1430
90 6.117570 64.233.169.104 71.192.34.104 HTTP 814 HTTP/1.1 200 OK (text/html)
91 6.118515 71.192.34.104 64.233.169.104 TCP 60 4335 → 80 [ACK] Seq=636 Ack=3621 Win=260176 Len=0
92 6.162091 169.254.247.145 169.254.255.255 NBNS 92 Name query NB HPAB9D4C<00>
93 6.241357 71.192.34.104 64.233.169.104 HTTP 719 GET /intl/en_ALL/images/logo.gif HTTP/1.1
94 6.273849 64.233.169.104 71.192.34.104 TCP 309 80 → 4335 [PSH, ACK] Seq=3621 Ack=1301 Win=8320 Len=0
95 6.274230 64.233.169.104 71.192.34.104 TCP 1484 80 → 4335 [ACK] Seq=3876 Ack=1301 Win=8320 Len=0

> Frame 85: 689 bytes on wire (5512 bits), 689 bytes captured (5512 bits)
> Ethernet II, Src: Dell_4f:36:23 (00:08:74:4f:36:23), Dst: Cisco_bf:6c:01 (00:0e:d6:bf:6c:01)
> Internet Protocol Version 4, Src: 71.192.34.104, Dst: 64.233.169.104
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 675
  Identification: 0xa2ac (41644)
  > 010. .... = Flags: 0x2, Don't fragment
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 127
  Protocol: TCP (6)
  Header Checksum: 0x022f [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 71.192.34.104
  Destination Address: 64.233.169.104
  [Stream index: 7]
  > Transmission Control Protocol, Src Port: 4335, Dst Port: 80, Seq: 1, Ack: 1, Len: 635
    Source Port: 4335
    Destination Port: 80
    [Stream index: 2]

```

8. In the NAT_ISP_side trace file, at what time is the first 200 OK HTTP message received from the Google server? What are the source and destination IP addresses and TCP source and destination ports on the IP datagram carrying this HTTP 200 OK message? Which of these fields are the same, and which are different than your answer to question 4 above?

At the time 6.117570

IP: Src: 64.233.169.104, Dst: 71.192.34.104

TCP in IP datagram: Src Port: 80, Dst Port: 4335

The time and the destination IP address are different; The source IP address, the TCP source port, and the destination port are the same.


```

80 6.117407 64.233.169.104 71.192.34.104 TCP 1484 80 → 4335 [ACK] Seq=1431 Ack=636 Win=7040 Len=1430 [TCP
90 6.117579 64.233.169.104 71.192.34.104 HTTP 814 HTTP/1.1 200 OK (text/html)
91 6.118515 71.192.34.104 64.233.169.104 TCP 60 4335 → 80 [ACK] Seq=636 Ack=3621 Win=260176 Len=0
92 6.162091 169.254.247.145 169.254.255.255 NNS 92 Name query NB HPAB9D4C<00>
93 6.241357 71.192.34.104 64.233.169.104 HTTP 719 GET /intl/en_ALL/images/logo.gif HTTP/1.1
94 6.273849 64.233.169.104 71.192.34.104 TCP 309 80 → 4335 [PSH, ACK] Seq=3621 Ack=1301 Win=8320 Len=255
95 6.274230 64.233.169.104 71.192.34.104 TCP 1484 80 → 4335 [ACK] Seq=3876 Ack=1301 Win=8320 Len=1430 [TCP
96 6.274571 64.233.169.104 71.192.34.104 TCP 1484 80 → 4335 [ACK] Seq=636 Ack=3621 Win=8320 Len=1430 [TCP

> Frame 90: 814 bytes on wire (6512 bits), 814 bytes captured (6512 bits)
> Ethernet II, Src: Cisco_bf:6c:01 (00:0e:d6:bf:6c:01), Dst: Dell_4f:36:23 (00:08:74:4f:36:23)
> Internet Protocol Version 4, Src: 64.233.169.104, Dst: 71.192.34.104
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x20 (DSCP: CS1, ECN: Not-ECT)
    Total Length: 800
    Identification: 0xf61e (63006)
  > 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 51
    Protocol: TCP (6)
    Header Checksum: 0x3a20 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 64.233.169.104
    Destination Address: 71.192.34.104
    [Stream index: 7]
  > Transmission Control Protocol, Src Port: 80, Dst Port: 4335, Seq: 2861, Ack: 636, Len: 760
    Source Port: 80
    Destination Port: 4335
    [Stream index: 2]
    > [Conversation completeness: Incomplete, DATA (15)]
    [TCP Segment Len: 760]
    Sequence Number: 2861 (relative sequence number)
    Sequence Number (raw): 3914286017
    [Next Sequence Number: 3621 (relative sequence number)]
    Acknowledgment Number: 636 (relative ack number)
    Acknowledgment number (raw): 4164041056
    0101 .... = Header Length: 20 bytes (5)
  
```

9. In the NAT_ISP_side trace file, at what time were the client-to-server TCP SYN segment and the server-to-client TCP ACK segment corresponding to the segments in question 5 above captured? What are the source and destination IP addresses and source and destination ports for these two segments? Which of these fields are the same, and which are different than your answer to question 5 above?

SYN:

At the time 6.035475

IP: Src: 71.192.34.104, Dst: 64.233.169.104

TCP: Src Port: 4335, Dst Port: 80

```

82 6.035475 71.192.34.104 64.233.169.104 TCP 66 4335 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=4 SACK_PERM
83 6.069799 64.233.169.104 71.192.34.104 TCP 66 80 → 4335 [SYN, ACK] Seq=0 Ack=1 Win=5720 Len=0 MSS=1430 SACK_PERM WS=64
84 6.068754 71.192.34.104 64.233.169.104 TCP 60 4335 → 80 [ACK] Seq=1 Ack=1 Win=260176 Len=0
85 6.069168 71.192.34.104 64.233.169.104 HTTP 689 GET / HTTP/1.1
86 6.092755 Cisco_bf:6c:01 Broadcast ARP 60 Who has 71.192.35.144? Tell 71.192.32.1
87 6.099637 64.233.169.104 71.192.34.104 TCP 60 80 → 4335 [ACK] Seq=1 Ack=636 Win=7040 Len=0
88 6.117078 64.233.169.104 71.192.34.104 TCP 1484 80 → 4335 [ACK] Seq=1 Ack=636 Win=7040 Len=1430 [TCP PDU reassembled in 9
89 6.117407 64.233.169.104 71.192.34.104 TCP 1484 80 → 4335 [ACK] Seq=1431 Ack=636 Win=7040 Len=1430 [TCP PDU reassembled in 9

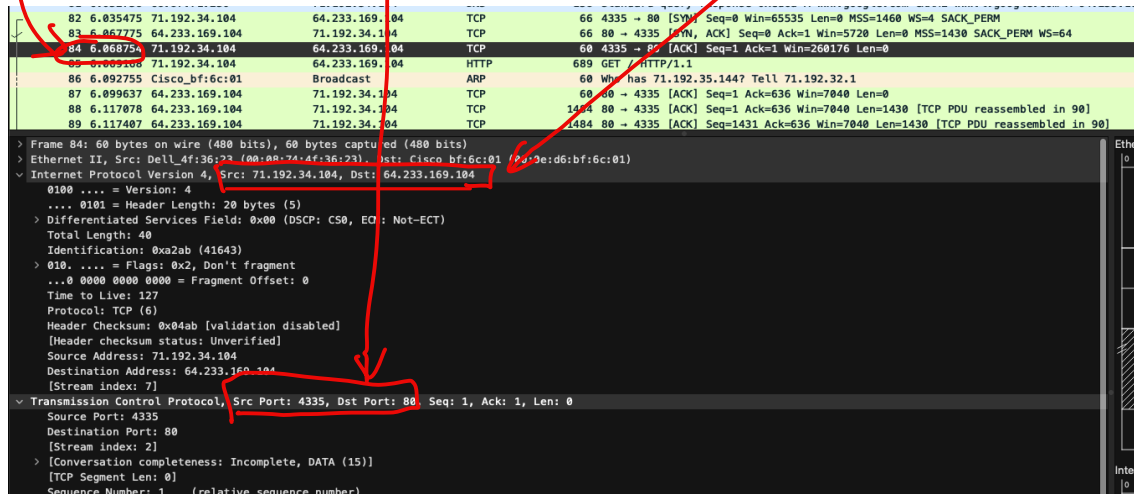
> Frame 82: 66 bytes on wire (528 bits), 66 bytes captured (528 bits)
> Ethernet II, Src: Dell_4f:36:23 (00:08:74:4f:36:23), Dst: Cisco_bf:6c:01 (00:0e:d6:bf:6c:01)
> Internet Protocol Version 4, Src: 71.192.34.104, Dst: 64.233.169.104
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 52
    Identification: 0xa2aa (41642)
  > 010. .... = Flags: 0x2, Don't fragment
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 127
    Protocol: TCP (6)
    Header Checksum: 0x04a0 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 71.192.34.104
    Destination Address: 64.233.169.104
    [Stream index: 7]
  > Transmission Control Protocol, Src Port: 4335, Dst Port: 80, Seq: 0, Len: 0
    Source Port: 4335
    Destination Port: 80
  
```

ACK:

At the time 6.068754

IP address: Src: 71.192.34.104, Dst: 64.233.169.104

TCP: Src Port: 4335, Dst Port: 80



For the SYN packet, only the source IP address is different, and the destination IP address, the TCP destination port, and source port number are the same.

For the ACK packet, only the source IP address has changed, and the destination IP address, the TCP destination and source port number are the same.

Thus, only the source IP addresses are different, the rest of it is the same.